

DATA 211 PROJECT

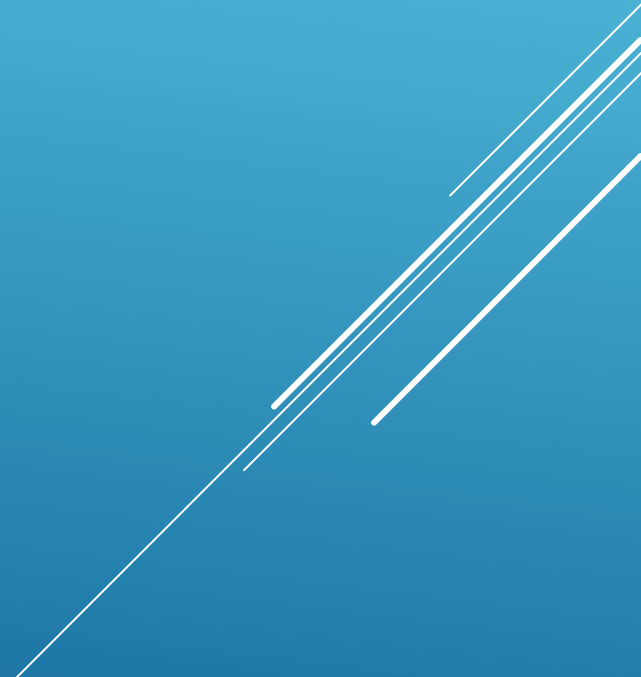
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Q?:

What affects the number of sleep I get in a day?

► Is it affected by whether it is a weekday or weekend?

RESEARCH QUESTION

Several thin, parallel white lines of varying lengths and slopes are positioned in the bottom right corner of the slide, creating a modern, abstract graphic element.

CLAIM

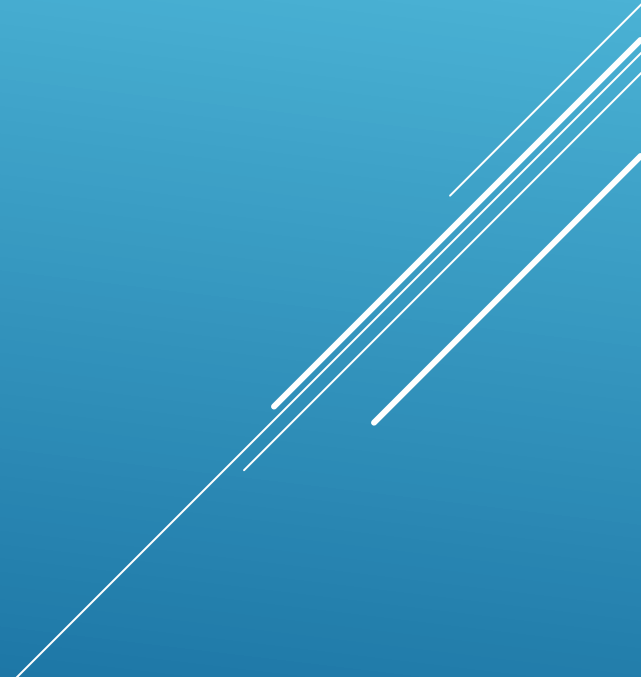
The average hours of sleep I get on a weekend day is different than the average hours of sleep I get on a weekday.

Several thin, white, parallel diagonal lines are located in the bottom right corner of the slide, extending from the right edge towards the center.

Independent Variable: The type of day (weekday or weekend)

Dependent Variable: The hours of sleep

VARIABLES

Several thin, white, parallel diagonal lines are positioned in the bottom right corner of the slide, extending from the right edge towards the center.

- 14 observations
- Tracked manually using start and end time

IV: Type of Day

DV: Hours of Sleep

DATA

	DayofWeek	Hours of Sleep	TypeofDay
1	Monday	7.1	Weekday
2	Tuesday	6.7	Weekday
3	Wednesday	7.3	Weekday
4	Thursday	7.4	Weekday
5	Friday	7.8	Weekday
6	Saturday	8.3	Weekend
7	Sunday	9.0	Weekend
8	Monday	6.6	Weekday
9	Tuesday	7.2	Weekday
10	Wednesday	7.5	Weekday
11	Thursday	8.0	Weekday
12	Friday	8.3	Weekday
13	Saturday	9.2	Weekend
14	Sunday	8.6	Weekend

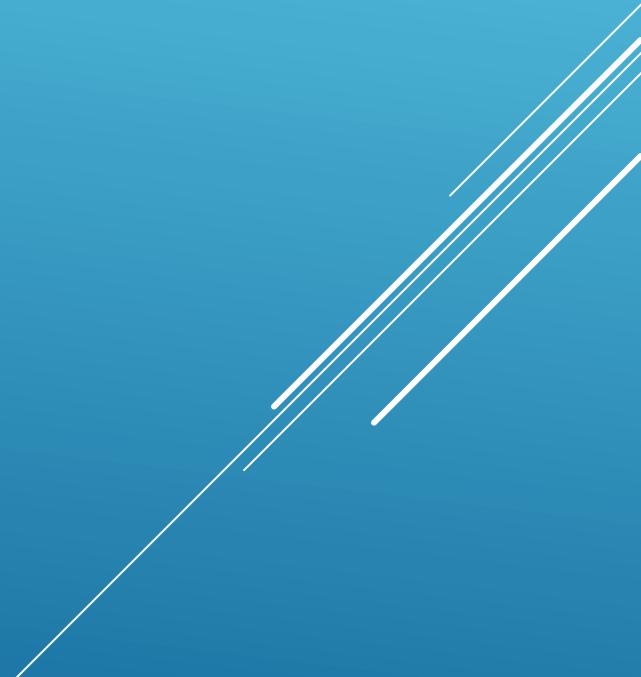
Step 1:

Establish Null and Alternative Hypotheses

Step 2:

Use T-test

HYPOTHESIS TESTING

A series of white diagonal lines of varying lengths and thicknesses, located in the bottom right corner of the slide.

Null Hypothesis:

Average hours of sleep on weekdays and weekends are the equal

Alternative Hypothesis:

Average hours of sleep on weekdays and weekends are different

HYPOTHESES

Significance Level = 0.05

p-value = 0.0009406

T-TEST

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Welch Two Sample t-test

data:  datapro$HoursofSleep by datapro$TypeofDay
t = -5.2493, df = 7.5308, p-value = 0.0009406
alternative hypothesis: true difference in means between group Weekday and group
Weekend is not equal to 0
95 percent confidence interval:
 -2.0000883 -0.7699117
sample estimates:
mean in group Weekday mean in group Weekend
      7.390             8.775
```




Since the p-value is less than the 0.05 significance level, the null hypothesis is rejected. There is enough evidence to support that the average hours of sleep on the weekday and weekend are different.

CONCLUSION